Week 4

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **2.** Compute and interpret z-scores to compare values on different scales or in different groups. (CLO 2) **3.** Find the five-number summary, range, and IQR of a dataset. (CLO 2) **4.** Apply the 1.5 × IQR rule to identify outliers. (CLO 2) **5.** Interpret a boxplot to describe a distribution's center, spread, skewness, and outliers. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2 Participation: Using Percentiles Your Turn: Comparing Across Groups (heights of men and women with z-scores) Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Kahoot! Zoom Respondus
6. Compare a quantitative variable across groups using side-by-side boxplots. (CLO 2) **7.** Compute and interpret a difference in means. (CLO 2) **8.** Describe the direction, form, and strength of a relationship in a scatterplot. (CLO 7) **9.** Estimate the correlation r from a scatterplot, and compute it using technology. (CLO 7) **10.** Explain how outliers and nonlinear patterns affect r. (CLO 7) **11.** Explain why correlation doesn't imply causation. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Your Turn: Tea vs. Coffee: Descriptive Statistics (difference in means; can we conclude causation?)	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM) Take-Home Quiz 3 (due Tue Oct 6, 11:59 PM) – covers 2.4–2.5 (objectives 8–11) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS Blackboard Zoom Respondus